TIPA Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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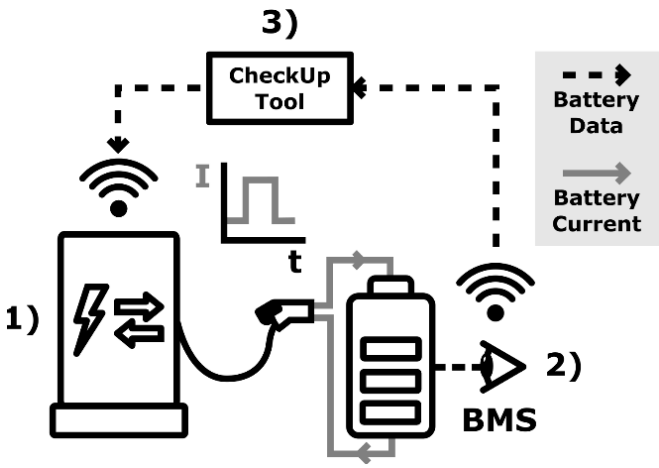
2. Technology Overview

Title of Proposed Technology	Algorithm development to estimate SoX and degradation from fast-charging BMS data for safe second-life decisions			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Electrical & Electronics		Electrical & Electronics	
	Medium Category	Small Category	Medium Category	Small Category
	Instrumentation	Safety monitoring/diagnosis measurement and control instruments	Batteries	Secondary Batteries
Concept of Proposed Technology	Fast charging profiles can contain rich information about battery condition, but they also introduce fast transients, temperature gradients, and strong non-linearities that degrade the accuracy of conventional BMS estimators. This proposal targets robust estimation of key State-of-X metrics (SoC, SoH, thermal state and power capability) and degradation indicators from fast-charging BMS signals, with quantified uncertainty, to improve second-life grading and			

	remanufacturing decisions. Core technical approach (model-based plus data-driven): • Physics-informed battery pack models (electrical, thermal, ageing) calibrated from cell and pack characterization data, and implemented in MATLAB/Simulink and in TNO's Battery Modular Simulation Tool (BMST) Simulink library [4]. • Model-based observers for SoX (e.g. Kalman-filter family of estimators and joint parameter identification) and thermal state observers designed for real-time BMS use [4][8][9]. • Temperature-aware modelling and identification methods to retain accuracy across cold and mild temperatures relevant to automotive operation [5][7]. • Data-driven fast-charge signatures and diagnostic indicators coupled to the model states to improve robustness to sensor bias, missing data, and varying duty cycles [4]. 
Rationale for Joint R&D Project	Reliable second-life and recycling decisions for lithium-ion batteries increasingly depend on accurate interpretation of in-use BMS and fast-charging data, as physical end-of-life testing is costly, slow, and not scalable. Fast charging, however, introduces strong non-linear electro-thermal effects and accelerates ageing, which increases uncertainty in estimating SoX (SoC, SoH), degradation, and remaining useful life using conventional BMS methods alone [4–9]. This creates a clear need for advanced, physics-informed diagnostic algorithms that are robust to temperature variation, measurement bias, and real-world operating conditions [5,7,9]. Korean SME contributes industrial expertise in battery reuse, remanufacturing, and second-life decision workflows, along with access to relevant operational BMS datasets and a clear commercialization pathway. TNO (Netherlands) complements this with internationally validated capabilities in battery modeling, observer-based SoX estimation, degradation analysis, and controlled validation using simulation (MIL/HIL) and laboratory testing infrastructure [4,7–11]. The joint R&D combines the Korean SME's market-driven requirements and deployment context with TNO's algorithmic depth and validation capability, reducing technical risk and accelerating progression toward deployable, trustworthy diagnostics for battery second-life grading and

	lifecycle services [4,6,11]
Key Words	1. SoX estimation 2. Battery degradation 3. Fast charging 4. BMS data analytics 5. Second-life valuation

3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6)
R&D Objectives	• Build a validated SoX and degradation estimation pipeline using fast-charging BMS data. • Develop models and indicators that are robust to BMS measurement errors and varying operating conditions. • Demonstrate decision support outputs for second-life applications: grading, remaining value, and safety-related flags. • Deliver a prototype software implementation and validation report suitable for integration into the SME's workflows.			
R&D Target Specifications	Target specifications (to be finalized during Phase 1 based on available data and use cases): • SoX estimation accuracy targets (example): SoH and SoC error within defined bounds agreed with Korean SME • Degradation tracking stability across diverse fast-charge sessions and temperatures • Model inference runtime suitable for batch processing and near real-time evaluation (requirements to be defined) • Explainability outputs to support auditability of grading decisions (features and evidence for key decisions) • Validation on representative datasets and, where available, linked ground truth tests (capacity, DCIR, or other lab metrics)			
R&D Plan	Proposed 36-month work plan (timeline and milestones to be aligned with the selected TIPA track): **WP1 Data definition and ground truth (Months 1-6)** - Define required channels from BMS and charger (current, voltage, temperature, timestamps, charge protocol metadata). - Data quality checks, synchronization, and anonymization; define evaluation splits by temperature, C-rate and ageing level. **WP2 Physics-informed baseline estimators (Months 4-12)** - Build temperature-aware equivalent-circuit and OCV models and calibration workflow. - Implement baseline SoX estimators (for example EKF/UKF observers with online parameter identification) and establish error metrics. **WP3 Robust joint estimation and degradation indicators (Months 10-24)**			

	- Develop joint estimation of SoC, SoH, and thermal state using observer-based methods and uncertainty quantification. - Derive fast-charge diagnostic features (for example resistance growth and relaxation signatures) and fuse with model states. - Develop ageing-aware power capability constraints and degradation indicators relevant to second-life grading. **WP4 Controlled validation and pilot integration (Months 18-33)** - Validate against Korean SME remanufacturing decision cases and controlled test data, including climate-chamber and MIL/HIL campaigns available at TNO (4). - Package a prototype toolchain (API plus reporting) and integrate into Korean SME workflows. **WP5 Demonstration, IP and dissemination (Months 30-36)** - Joint demonstration and documentation; prepare at least one peer-reviewed dissemination output. - IP review and filing plan, where applicable; commercialization and scale-up plan for Korean SME services.

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	TNO (Research Institute) expected roles: • Technical Lead for model-based algorithm development for SoX and degradation estimation from fast-charge BMS data, including temperature-aware electrical models, thermal state observers, and ageing-aware indicators. • Provide validated modelling and simulation environment (MATLAB/Simulink and BMST Simulink library) and implement reference estimators for benchmarking (4). • Design validation protocol and run model-in-the-loop (MIL) and hardware-in-the-loop (HIL) verification for estimator robustness across temperature and duty-cycle conditions, leveraging TNO test environments (4). • Transfer algorithms to Korean SME in a form suitable for integration (documented Python and C/C++ deployable modules, plus performance and uncertainty reporting). • Co-author joint publications and support IP and commercialization strategy with the Korean SME partner.
Expected R&R of Korean SME Partner	Korean SME partner expected roles: • Provide requirements and representative use cases for second-life decision-making • Provide access to relevant BMS datasets, including fast-charging data, under agreed data governance • Provide domain expertise on pack remanufacturing processes and grading criteria • Support labeling and ground truth linkage where available (capacity, resistance, diagnostics) • Lead field validation and integration of outputs into the SME's battery lifecycle services

5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected outcomes: • A validated algorithmic toolkit for SoX and degradation estimation tailored to fast-charging BMS datasets • Demonstrated decision support outputs for second-life grading, remaining value estimation, and safety-relevant flags • Prototype software implementation and evaluation reports supporting deployment in Korean SME operations Expected impact: • Improved reliability and transparency of second-life battery reuse decisions, supporting higher-value reuse pathways • Reduced uncertainty and improved asset valuation for battery lifecycle services • Strengthened international competitiveness of Korean SME services and broader ecosystem readiness for battery lifecycle regulation

6. References

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale

	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization